

Capstone Project Write-Up

Portfolio Resilience: Backtesting Asset Allocation Strategies During Financial Crises

Dane Herrin

Alexander Jackson

2026-06-09

Table of contents

1	Title, Team, and Affiliations	2
2	Abstract	2
3	Introduction: Problem and Research Questions	2
4	Impact and Stakeholders	3
5	Related Work and Background	3
6	Data and Data Engineering	3
7	Ethics and Responsible Use	4
8	Methods and Evaluation	5
9	Results and Visualizations	5
10	Discussion, Limitations, and Future Work	5
11	Conclusions and Recommendations	6
12	Reproducibility and References	6
13	References	6

1 Title, Team, and Affiliations

Project title: *(same as charter)*

Name	Role	Willamette email
	Dane Herrin	dmherrin@willamette.edu
	Alex Jackson	

Peer Stakeholder POs: *(from lookup)*

Repository: <https://github.com/your-org/data510--Quant-Back-Testing->

2 Abstract

(150 to 250 words after you have results. Summarize: problem, approach, key finding, impact, limitation. Write this section last.)

3 Introduction: Problem and Research Questions

Many stakeholders evaluate portfolios based on expected returns, recent performance, or exposure to popular sectors. However, financial crises often reveal risks that are not obvious during normal market conditions. Portfolios that appear strong during growth years may become fragile when volatility rises, correlations shift, and drawdowns accelerate. This project focuses on backtesting different portfolios across periods of financial stress. The goal is to evaluate how different sectors and asset classes behave during crisis periods compared to normal market conditions. The project will compare return potential, downside protection, volatility, recovery behavior, and risk-adjusted performance across multiple portfolio strategies.

Problem statement: Stakeholders often construct or evaluate portfolios based on expected gains without fully understanding how those portfolios behave during financial crises and periods of market uncertainty.

Primary research question: How can portfolio optimization help stakeholders construct portfolios that preserve return potential while reducing drawdowns during periods of financial crisis and market uncertainty?

Secondary questions (if any): Do drawdown-resistant portfolios experience significantly lower gains during growth years compared to regular market indexes? How does portfolio

performance during crisis periods compare to performance during normal market conditions? How are Sharpe and Sortino ratios affected during periods of uncertainty compared to periods of growth? How do different asset classes and sectors, such as technology, energy, real estate, consumer staples

Changes since M1 proposal: *none yet*

4 Impact and Stakeholders

The main stakeholders include: * Investors * Portfolio managers * Financial analysts * Finance educators * Investment clubs * Risk managers * Students studying finance or portfolio analytics

- If successful, this project will help stakeholders understand how portfolio construction choices affect performance during both growth periods and crisis periods. The project will show how drawdowns, volatility, recovery time, Sharpe ratio, and Sortino ratio change across different market environments.

5 Related Work and Background

Wildfire smoke and respiratory health are well studied at regional scale (Reid et al. 2016; Liu et al. 2016). This project applies county-level public data to Oregon specifically.

6 Data and Data Engineering

- The project will use a combination of individual equities, sector ETFs, and benchmark ETFs.
- Potential asset groups include:
 - Technology: Apple, Microsoft, Nvidia, Amazon, Google, Meta, Broadcom, Tesla
 - Energy: ExxonMobil, Chevron, ConocoPhillips, EOG Resources, Schlumberger, utilities, and energy-related holdings
 - Real estate: VNQ or similar REIT ETF
 - Consumer staples: Walmart, Costco, Target, Procter & Gamble, Coca-Cola, PepsiCo, General Mills, and similar defensive consumer names
 - Defensive assets: Treasury ETF or bond ETF
 - Benchmarks: SPY, QQQ, BND, Fama-French Rm-Rf market composite

Sources:

Source	Role	Access
--------	------	--------

Engineering summary: Raw data in `data/raw/`; processed panel in `data/processed/`; ingest scripts in `src/ingest/`.

```
# Replace with your project data when ready:
# panel <- readRDS(here::here("data/processed/analysis_panel.rds"))
# nrow(panel)
knitr::kable(data.frame(
  check = c("Rows", "Counties", "Date range"),
  status = c("stub", "stub", "stub")
))
```

Table 3: Example QC summary (replace with your panel)

check	status
Rows	stub
Counties	stub
Date range	stub

7 Ethics and Responsible Use

- **This project could be misread as recommending specific investments. To avoid that, all results will be framed as historical analysis, not financial advice. The project will clearly state that past performance does not guarantee future results.**
- **Privacy:** All data used is free and open source with no identifiable ethical concerns.
- **Fairness / equity:** All data used is available to everyone and as fairly represented as possible.
- **Limitations of setting:** Backtesting can create false confidence if portfolios are built using knowledge of what performed well historically. This can create a survivorship bias.
- **Non-causal framing:** We will address this by documenting portfolio construction rules before evaluation and avoiding claims that the results will hold in future crises.

8 Methods and Evaluation

Design:

Primary analysis:

Evaluation metrics:

Train / validation / holdout:

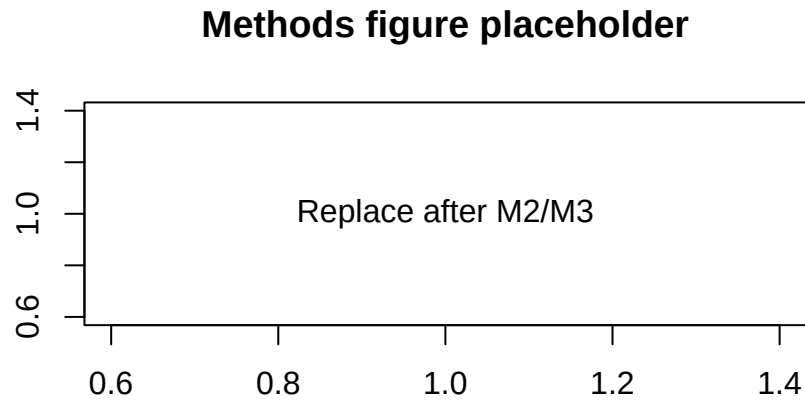


Figure 1: Replace with a methods diagram or exploratory plot

9 Results and Visualizations

(Stub: key findings with Figure 1 and tables.)

10 Discussion, Limitations, and Future Work

Interpretation:

Limitations: *(tie to actual data and design)*

Future work:

11 Conclusions and Recommendations

(Short, stakeholder-facing takeaways.)

12 Reproducibility and References

Reproduce main results:

```
cd your-repo
quarto render deliverables/M4-writeup-draft/writeup.qmd
```

Dependencies: `requirements.txt` or `renv.lock` in repo root.

Data: URLs and approval status documented in `docs/data-dictionary.md`.

13 References

- Liu, J. C., L. J. Mickley, M. P. Sulprizio, et al. 2016. “Wildfire-Specific Fine Particulate Matter and Risk of Hospital Admissions in Urban and Rural Counties.” *Epidemiology* 27 (3): 350–57.
- Reid, C. E., M. Brauer, F. H. Johnston, et al. 2016. “Critical Review of Health Impacts of Wildfire Smoke Exposure.” *Environmental Health Perspectives* 124 (8): 1334–43.